

Computer Science & Information Technology

C - Programming

Data Types & Operators

DPP: 3

Q1 include <stdio.h>

```
int main(void)
{
    int a;
    a = 2 * 6/5 + 3.0/2 + 1;
    printf("%d", a);
    return 0;
}
```

The value of a is ____

- (A) 4.9 (B) 4.0
(C) 4.5 (D) 4

Q2 #include <stdio.h>

```
int main(void)
{
    int a;
    a = 16.0 /4 * 5 % 3;
    printf("%d", a);
    return 0;
}
```

The value of a printed is ____

- (A) Compiler error (B) 8.0
(C) 2 (D) 8

Q3 Consider the following program.

```
#include<stdio.h>
void main()
{
    int a;
    a=32>24>13>10>8>-1>0;
    printf("%d",a);
}
```

The output is

Q4 #include<stdio.h>

```
void main()
{
    int a;
```

```
a=25>15>0!=12<45>42!= 65;
```

```
printf("%d",a);
```

```
}
```

The output is ____.

Q5 Consider the following program:

```
#include<stdio.h>
void main()
{
    int a=0, b=1;
    a=(a=5)&&(b=0);
    printf("%d", a);
    printf("%d", b);
}
```

The output is:

- (A) 50 (B) 00
(C) 10 (D) Compiler error

Q6 Consider the following statements:

P: The precedence of the modulus operator is higher than multiplication or division operator.

Q: The result of the modulus operator contains the sign of the second operand.

Which of the following statements is/are INCORRECT?

- (A) Only P
(B) Only Q
(C) Both P and Q
(D) Neither P nor Q

Q7 Consider the following program:

```
#include<stdio.h>
void main()
{
    int a=2022;
    printf("%d%d%d", a!=2024, a=2023,
a==2021);
}
```


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The output is-

- (A) 020220 (B) 020231
(C) 002021 (D) 120230

Q8 Consider the following program:

```
#include<stdio.h>
```

```
void main()
```

```
{  
    int x=-2023;  
    printf("%d", ~(x=x+5));  
}
```

The output is _____.



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Answer Key

Q1 (D)

Q2 (A)

Q3 1~1

Q4 1~1

Q5 (B)

Q6 (C)

Q7 (D)

Q8 2017~2017



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Hints & Solutions

Q1 Text Solution:

$$\begin{aligned} a &= 12/5 + 3.0/2 + 1 \\ &= 2 + 1.5 + 1 \\ &= 4.5 \end{aligned}$$

a is an integer, so a = 4

Q2 Text Solution:

$$\begin{aligned} a &= 16.0/4 * 5 \% 1 \\ &= 4.0 * 5 \% 3 \\ &= 20.0 \% 3 \\ &= \text{Error} \end{aligned}$$

Modulus operator works only with integers.

Q3 Text Solution:

$$\begin{aligned} a &= 32 > 24 > 13 > 10 > 8 > -1 > 0 \\ &\quad 1 > 13 \text{ } 0 > 10 \\ &\quad \quad 0 > 8 \\ &\quad \quad \quad 0 > -1 \\ &\quad \quad \quad \quad 1 > 0 \\ &\quad \quad \quad \quad \quad 1 \end{aligned}$$

Q4 Text Solution:

$$\begin{aligned} a &= 25 > 15 > 0! = 12 < 45 > 42! = 65 \\ &\quad 1 > 0 \\ &\quad \quad 1! = 12 \\ &\quad \quad \quad 1 < 45 \\ &\quad \quad \quad \quad 1 > 42 \text{ } 0! = 65 \\ &\quad \quad \quad \quad \quad 1 \\ &\quad \quad \quad \quad \quad a = 1; \end{aligned}$$

output : 1

Q5 Text Solution:

```
int a=0, b=1;
a=(a=5)&&(b=0);
// Assignment operator assigns and returns the
assigned value. So, a=5&&0=0, b=0
printf("%d", a);//0 is printed
printf("%d", b);//0 is printed.
```

Q6 Text Solution:

P: INCORRECT. The precedence of the modulus operator is same as multiplication or division operator.

Q: INCORRECT. The result of the modulus operator contains the sign of the first operand.

Q7 Text Solution:

a=2022. So, a!=2024 evaluates to 1.
a=2023. Assignment operator assigns the value and returns the assigned value.
a==2021 is equivalent to 2023==2021. So, it evaluates to 0.
Output: 120230

Q8 Text Solution:

$x=x+5 \rightarrow x=-2023+5=-2018$
 $\sim(x) \rightarrow \sim(-2018) = -(-2018+1) = 2017$.
Output: 2017.



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